

KUDAT, Malaysia

WMO: 964770

Lat: 6.917N

Lon: 116.833E

Elev: 16

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	73.0	73.2	69.5	108.6	81.4	71.2	114.9	79.8	17.0	82.0	15.7	82.1	2.7	230	0.348

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	12.5	92.9	80.3	91.6	80.2	91.1	80.1	82.8	89.3	81.9	88.5	81.5	87.8	9.0	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.9	160.7	87.3	80.4	158.2	87.1	79.2	151.6	85.5	46.6	89.9	45.7	89.7	45.2	89.2	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.5	13.6	12.0	70.0	94.6	1.5	1.4	68.9	95.6	68.0	96.3	67.1	97.1	66.0	98.1		
(5)			69.3	84.3	1.3	1.4	68.4	85.3	67.6	86.1	66.9	86.9	66.0	87.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	82.0	80.7	81.3	82.2	83.5	83.6	82.7	82.0	82.3	82.0	81.7	81.4	81.0	(6)
(7)		DBStd	2.05	2.00	2.10	1.98	2.06	1.85	1.71	1.73	1.72	1.69	1.75	1.64	1.98	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	11689	950	876	997	1004	1042	982	992	1000	960	983	941	962	(10)
(11)		CDD65	6214	485	456	532	554	577	532	527	535	510	518	491	497	(11)
(12)		CDH74	65633	4693	4598	5763	6571	6651	5805	5414	5717	5294	5271	4946	4910	(12)
(13)		CDH80	23545	1144	1212	1938	2764	2874	2391	2062	2311	1986	1855	1617	1391	(13)
(14)	Wind	WSAvg	5.3	6.1	6.6	5.7	4.9	4.6	4.7	5.1	5.3	5.2	5.4	4.5	5.1	(14)
(15)	Precipitation	PrecAvg	96.9	16.5	8.4	6.7	3.0	5.4	5.6	5.0	4.7	4.6	7.3	10.5	19.1	(15)
(16)		PrecMax	158.0	39.6	27.3	20.1	5.9	10.8	8.9	11.6	7.8	7.8	14.2	19.7	35.8	(16)
(17)		PrecMin	59.4	4.8	1.7	1.4	0.2	1.3	0.7	0.7	1.1	0.9	4.0	2.2	5.0	(17)
(18)		PrecStd	24.8	9.4	6.9	5.3	1.7	2.6	2.1	2.5	2.0	1.9	2.7	4.2	8.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.2	89.7	91.0	93.4	93.6	93.4	91.8	93.2	91.8	91.5	91.0	89.9	(19)
(20)			MCWB	80.0	80.1	80.8	82.0	81.4	79.6	79.2	78.5	79.1	80.2	81.1	80.5	(20)
(21)		2%	DB	87.7	88.0	89.6	91.7	91.8	91.7	91.3	91.5	91.1	89.8	89.5	88.1	(21)
(22)			MCWB	79.8	79.7	80.1	81.3	81.0	79.7	79.3	79.0	79.4	79.8	80.2	80.2	(22)
(23)		5%	DB	86.2	86.4	88.1	91.0	91.3	91.0	89.7	90.0	89.6	89.3	88.0	87.4	(23)
(24)			MCWB	79.1	78.9	79.9	81.0	80.8	79.8	79.0	79.0	79.2	79.8	80.1	80.0	(24)
(25)		10%	DB	84.6	85.8	87.5	89.5	89.8	89.5	88.1	89.4	88.1	87.8	86.4	85.9	(25)
(26)			MCWB	78.4	78.6	79.7	80.4	80.5	79.9	79.2	79.0	79.4	79.8	79.5	79.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.9	81.0	82.7	83.5	83.7	84.3	82.2	82.4	81.9	82.2	82.0	81.9	(27)
(28)			MCD B	86.0	86.7	87.9	90.9	90.5	89.0	89.4	89.5	89.0	88.9	87.7	87.1	(28)
(29)		2%	WB	80.2	80.3	81.2	82.3	82.4	82.4	81.0	81.2	81.0	81.1	81.0	80.9	(29)
(30)			MCD B	84.9	85.9	86.9	89.4	89.4	88.7	87.9	88.1	87.7	87.3	86.4	85.6	(30)
(31)		5%	WB	79.8	79.8	80.4	81.6	81.6	81.2	80.1	80.2	80.1	80.4	80.5	80.3	(31)
(32)			MCD B	84.2	84.9	86.1	88.1	88.3	87.6	86.7	86.9	86.4	86.4	85.6	84.7	(32)
(33)		10%	WB	79.2	79.1	79.8	80.9	81.0	80.3	79.7	79.7	79.7	79.9	80.0	79.8	(33)
(34)			MCD B	83.3	83.7	85.1	86.9	87.4	86.7	86.0	86.4	85.9	85.6	84.9	84.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.8	8.9	10.7	12.5	13.7	14.2	13.6	14.4	13.4	12.4	11.5	9.5	(35)
(36)			MCDBR	10.2	10.1	11.9	14.0	14.8	16.1	15.1	15.7	15.1	14.0	13.1	11.7	(36)
(37)			MCWBR	5.0	4.7	5.2	5.8	6.0	6.6	6.2	6.4	6.1	6.0	6.0	5.5	(37)
(38)		5% WB	MCD BR	9.3	9.5	11.1	13.0	13.9	14.6	14.1	14.9	14.0	13.0	11.9	10.2	(38)
(39)			MCWBR	4.9	4.8	5.3	5.8	6.2	7.1	6.4	6.5	6.2	6.1	5.8	5.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.430	0.426	0.424	0.424	0.411	0.416	0.418	0.445	0.440	0.444	0.419	0.423	(40)	
(41)		taud	2.459	2.444	2.456	2.447	2.506	2.505	2.497	2.394	2.404	2.394	2.482	2.480	(41)	
(42)		Ebn at Noon	281	285	285	281	279	274	274	272	277	276	283	281	(42)	
(43)		Edh at Noon	36	37	37	37	34	34	34	39	39	39	35	35	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1404	1678	1828	1945	1791	1619	1596	1650	1619	1530	1430	1330	(44)	
(45)		RadStd	186	200	209	139	124	110	100	107	128	103	125	158	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	-0.67	-0.67	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	+0.65	+0.70	+0.54	N/A	N/A	+178	+176

Nomenclature: See separate page